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⑥ Solve the recurrence Relation

$$a_n = a_{n-1} + 3n^2 + 3n + 1 \text{ where } a_0 = 1$$

$$\text{Given, } a_n = a_{n-1} + 3n^2 + 3n + 1$$

$$\Rightarrow a_1 = a_0 + 3(1)^2 + 3(1) + 1 \\ = 1 + 3 + 3 + 1 = 8 = 2^3$$

$$\Rightarrow a_2 = a_1 + 3(2)^2 + 3(2) + 1 \\ = 8 + 12 + 6 + 1 = 27 = 3^3$$

$$\Rightarrow a_3 = a_2 + 3(3)^2 + 3(3) + 1 \\ = 27 + 27 + 9 + 1 = 64 = 4^3$$

$$\Rightarrow \boxed{a_n = (n+1)^3}$$

⑦

Solve the recurrence relation $F_n = 10F_{n-1} - 25F_{n-2}$

$$\text{Given } F_n = 10F_{n-1} - 25F_{n-2}$$

$= F_n - 10F_{n-1} + 25F_{n-2}$ which is an H.L.R.R of degree 2.

$$\text{char eqn} = c(m) = 0 \Rightarrow m^2 - 10m + 25 = 0.$$

$$(m-5)^2 = 0$$

$$\therefore m = 5, 5.$$

$$C.F = (c_1 + c_2 n) 5^n$$

As there are no conditions given,

$\therefore$ The required General Solution is $(c_1 + c_2 n) 5^n = a_n$

$$\boxed{a_n = (c_1 + c_2 n) 5^n}$$

Maths Assignment - IV

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Section : CSE - B

1 Q. Define Recurrence Relation.

Ans:- The Recurrence Relation for a sequence $\{a_0, a_1, a_2, \dots, a_n, \dots\}$ is the relation between n th term and its previous terms ~~and~~ for all integers $n \geq 0$

$$\text{Ex:- } a_n = a_{n-1} + 3a_{n-2}$$

2 Q. Define Fibonacci sequence and write its recurrence Relation.

Ans:- The fibonacci sequence is a series of numbers in which each number is the sum of two preceding numbers. The sequence's Recurrence Relation is,

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2, F_0 = 1, F_1 = 1$$

3 Q. Find the Particular solution for RR using the method of determined coefficients $a_n - 5a_{n-4} = 3^n$?

$$\text{Consider, } a_n - 5a_{n-4} = 0$$

$$\text{Characteristic eqn} = c(m) = 0 \Rightarrow m^4 - 5 = 0 \quad \text{--- (1)}$$

PI :- RHS of the given eqn is $1 \cdot 3^n$ where 3 is not a characteristic root as it is not satisfying (1).

$$\begin{aligned} a_n^p &= A \cdot 3^n \\ &= A \cdot 3^n - 5(A \cdot 3^{n-4}) \\ &= A \cdot 3^n - 5A \left(\frac{3^n}{3^4} \right) = 3^n \\ &= A \cdot \left(1 - \frac{5}{3^4} \right) = 1 \Rightarrow \boxed{A = \frac{81}{76}} \end{aligned}$$

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$$\therefore a_n^P = \frac{81}{36} \cdot 3^n \text{ is the required solution.}$$

(4) Find the generating function of the sequence $1, 2, 3, 4, \dots$?

Sol:

Given, $\{1, 2, 3, 4, \dots\}$

$$\begin{aligned} A(x) &= 1 + 2x + 3x^2 + 4x^3 + \dots \\ &= (1-x)^{-2} \quad (\text{or}) \quad \sum_{x=0}^{\infty} (x+1)x^2 \end{aligned}$$

(5) Solve $a_{n+2} - 5a_{n+1} + 4a_n = 0$ for $n \geq 0$ given $a_0 = 0, a_1 = 13$

Given, $a_{n+2} - 5a_{n+1} + 4a_n = 0$ of degree 2.

$$\text{char eqn} = c(m) = 0 \Rightarrow m^2 - 5m + 4 = 0.$$

$$m^2 - 4m - m + 4 = 0$$

$$(m-1)(m-4) = 0$$

$$m = 1, 4$$

$$\therefore \text{CF} = a_n^H = c_1 1^n + c_2 4^n$$

$$a_n = c_1 1^n + c_2 4^n$$

$$\text{Given, } a_0 = 0 \Rightarrow c_1 1^0 + c_2 4^0 = 0$$

$$\Rightarrow c_1 + c_2 = 0 \quad \text{--- (1)}$$

$$\text{Given } a_1 = 13 \Rightarrow c_1 1^1 + c_2 4^1 = 13$$

$$c_1 + 4c_2 = 13 \quad \text{--- (2)}$$

$$(2) - (1) = c_1 + 4c_2 - c_1 - c_2 = 13$$

$$3c_2 = 13$$

$$\Rightarrow c_1 = 0 - \frac{13}{3} = -\frac{13}{3}$$

$$c_2 = \frac{13}{3}$$

$$\therefore \text{The required solution} = a_n = \left(-\frac{13}{3}\right) 1^n + \left(\frac{13}{3}\right) 4^n$$

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⑧ Find Particular solution using the method of undetermined coefficients. $a_n - 3a_{n-1} + 2a_{n-2} = 5n + 3$

Consider, $a_n - 3a_{n-1} + 2a_{n-2}$ which is H.L.R.R of degree 2.

$$\text{Characteristic eqn} = c(m) = 0.$$

$$\therefore m^2 - 3m + 2 = 0.$$

$$m^2 - 2m - m + 2 = 0 \Rightarrow m(m-2) - 1(m-2) = 0.$$

$$(m-1)(m-2) = 0 \quad \therefore m = 1, 2.$$

$$\text{C.F.} = a_n^H = c_1 1^n + c_2 2^n$$

P.I = RHS of given eqn = $(5n+3) \cdot 1^n$ where 1^n is

char root of multiplicity 1 so,

$$a_n^P = (A_0 + A_1 n) \cdot 1^n \cdot n^1 \Rightarrow A_0 n + A_1 n^2$$

$$\Rightarrow A_0 n + A_1 n^2 - 3[A_0(n-1) + A_1(n-1)^2] + 2[A_0(n-2) + A_1(n-2)^2] = 5n + 3$$

$$\Rightarrow A_0 n + A_1 n^2 - 3A_0 n + 3A_0 - 3A_1(n^2 - 2n + 1) + 2A_0 n - 4A_0 + 2A_1(n^2 - 4n + 4) = 5n + 3$$

Compare the coeff of n :-

$$\Rightarrow A_0 - 3A_0 + 6A_1 + 2A_0 - 8A_1 = 5$$

$$\Rightarrow -2A_1 = 5 \Rightarrow \boxed{A_1 = -5/2}$$

Compare constants (put $n=0$):

$$\Rightarrow 3A_0 - 3A_1 - 4A_0 + 8A_1 = 3$$

$$\Rightarrow -A_0 + 5A_1 = 3$$

$$\Rightarrow -A_0 - \frac{25}{2} = 3 \Rightarrow -A_0 = 3 + \frac{25}{2}$$

$$\Rightarrow \boxed{A_0 = -\frac{31}{2}}$$

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$$\therefore a_n^p = \left(-\frac{31}{2} - \frac{5n}{2} \right) \cdot 1^n \cdot n$$

$\therefore$ The General Solution is

$$a_n = c_1 (1)^n + c_2 (2)^n - \left(\frac{31+5n}{2} \right) 1^n \cdot n$$

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Solve the recurrence relation.

$$a_k - 7a_{k-1} + 10a_{k-2} = 6 + 8k, \quad a(0) = 1, \quad a(1) = 2$$

Consider $a_k - 7a_{k-1} + 10a_{k-2}$ which is H.L.R.R of degree 2.

$$\text{Characteristic eqn} = c(m) = 0$$

$$m^2 - 7m + 10 = 0$$

$$m^2 - 5m - 2m + 10 = 0$$

$$m(m-5) - 2(m-5) = 0$$

$$(m-2)(m-5) = 0 \quad \therefore m = 2, 5$$

$$C.F = c_1 (2)^n + c_2 (5)^n = a_n^H$$

P.I :- RHS of given eqn = $(6+8k)1^k$ where, 1 is not char root. So,

$$a_k^p = (A_0 + A_1 k) 1^k = A_0 + A_1 k$$

$$\text{Consider, } a_k - 7a_{k-1} + 10a_{k-2} = 6 + 8k$$

$$\Rightarrow (A_0 + A_1 k) - 7(A_0 + A_1 (k-1)) + 10[A_0 + A_1 (k-2)] = 6 + 8k$$

$$\Rightarrow A_0 + A_1 k - 7A_0 + 7A_1 (k-1) + 10A_0 + 10A_1 (k-2) = 6 + 8k$$

$$\Rightarrow 4A_0 + 4A_1 k - 7A_1 = 6 + 8k$$

$$\Rightarrow 4A_0 + 4A_1 k - 7A_1 = 6 + 8k$$

$$\therefore 4A_1 = 8 \quad [\because \text{compare coeffs}]$$

$$\Rightarrow \boxed{A_1 = 2}$$

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compare constants. :-

$$4A_0 - 13A_1 = 6 \Rightarrow 4A_0 - 26 = 6$$

$$4A_0 = 32 \Rightarrow \boxed{A_0 = 8.}$$

$$\therefore a_k = c_1(2)^k + c_2(5)^k + (8+2k) \cdot 1^k.$$

$$\text{Given } a_0 = 1 \Rightarrow c_1 \cdot 2^0 + c_2 \cdot 5^0 + (8+0) \cdot 1^0$$

$$= c_1 + c_2 = -7 \quad \text{--- (1)}$$

$$a_1 = c_1 \cdot 2^1 + c_2 \cdot 5^1 + (8+2) \cdot 1^1$$

$$= 2c_1 + 5c_2 = -8 \quad \text{--- (2)}$$

$$\text{(2)} - 2 \times \text{(1)} = 2c_1 + 5c_2 - 2c_1 - 2c_2 = -8 + 14$$

$$3c_2 = 6 \Rightarrow c_2 = 2.$$

$$\text{From (1)} \quad c_1 + c_2 = -7.$$

$$\Rightarrow c_1 + 2 = -7 \Rightarrow c_1 = -9.$$

$$\therefore \boxed{a_k = -9 \cdot 2^k + 2 \cdot 5^k + (8+2k) \cdot 1^k} \text{ is the required}$$

solution.

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(a) Solve the recurrence relation.

$$a_n = 4a_{n-1} - 4a_{n-2} + (n+1)2^n.$$

Consider, $a_n - 4a_{n-1} + 4a_{n-2}$ is H.L.R.R of degree 2.

$$\text{Char eqn} = c(m) = 0.$$

$$\Rightarrow m^2 - 4m + 4 = 0.$$

$$m^2 - 2m - 2m + 4 = 0.$$

$$(m-2)(m-2) = 0$$

$$\therefore m = 2, 2.$$

$$\therefore \text{CF} = a_n^H = (c_1 + c_2 n) \cdot 2^n.$$

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P.I :- RHS of given eqn = $(n+1)2^n$ where 2 is char root of the char eqn.

$$\Rightarrow a_n^p = (A_0 + A_1 n) 2^n \cdot n^2$$

$$\text{Consider, } a_n - 4a_{n-1} + 4a_{n-2} = (n+1) \cdot 2^n.$$

$$\Rightarrow (A_0 + A_1 n) n^2 \cdot 2^n - 4[A_0 + A_1(n-1)(n-1)^2 \cdot 2^{n-1}] + 4[A_0 + A_1(n-2)] \cdot (n-2)^2 \cdot 2^{n-2} = (n+1)2^n$$

$$(n-2)^2 \cdot 2^{n-2} = (n+1)2^n$$

$$\Rightarrow (A_0 + A_1 n) \cdot n^2 \cdot 2^n - 4[A_0 + A_1(n-1)] \cdot (n^2 - 2n + 1) \frac{2^n}{2} + 4[A_0 + A_1(n-2)] \cdot (n^2 - 4n + 4) \frac{2^n}{4} = (n+1)2^n$$

$$\Rightarrow \cancel{(A_0 + A_1 n) n^2} - 2(A_0 + A_1(n-1)) \cdot (n^2 - 2n + 1) + (A_0 + A_1(n-2)) \cdot (n^2 - 4n + 4) = (n+1)2^n$$

$$\Rightarrow (A_0 + A_1 n) n^2 - 2(A_0 + A_1(n-1)) \cdot (n^2 - 2n + 1) + (A_0 + A_1(n-2)) \cdot (n^2 - 4n + 4) = (n+1)$$

$$\text{Compare coeff} \Rightarrow 4A_0 - 2A_1 - 4A_1 - 4A_0 + 4A_1 + 8A_1 = 1$$

$$\Rightarrow 6A_1 = 1 \Rightarrow \boxed{A_1 = 1/6}$$

$$\text{Compare constants :- } n=0 \text{ then } -2A_0 + 2A_1 + 4A_0 - 8A_1 = 1$$

$$\Rightarrow 2A_0 - 6A_1 = 1 \Rightarrow 2A_0 - 6\left(\frac{1}{6}\right) = 1$$

$$\Rightarrow 2A_0 = 2 \Rightarrow \boxed{A_0 = 1}$$

$$\therefore a_n^p = \left(1 + \frac{n}{6}\right) \cdot n^2 \cdot 2^n$$

$\therefore$ The required General Solution is :-

$$\boxed{(C_1 + C_2 n) 2^n + \left(1 + \frac{n}{6}\right) \cdot n^2 \cdot 2^n}$$

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b) $a_n = 9a_{n-1} - 27a_{n-2} + 27a_{n-3}$ with $a_0 = 5, a_1 = 24, a_2 = 117$

Consider, $a_n - 9a_{n-1} + 27a_{n-2} - 27a_{n-3} = 0$ which is

H.L.R.R of degree 3.

Char eqn = $c(m) = 0 \Rightarrow m^3 - 9m^2 + 27m - 27 = 0.$

$\Rightarrow m = 3, \quad m^2 - 6m + 9 = 0$

$(m-3)(m-3) = 0$

$m = 3, 3.$

$$\begin{array}{r|rrrr} & 1 & -9 & 27 & -27 \\ 3 & 0 & 3 & -18 & 27 \\ \hline & 1 & -6 & 9 & 0 \end{array}$$

$\Rightarrow m^2 - 6m + 9 = 0.$

$\therefore m = 3, 3, 3.$

$\therefore CF = a_n^N = (c_1 + c_2 n + c_3 n^2) 3^n.$

Given, $a_0 = 5 \Rightarrow (c_1 + c_2(0) + c_3(0)) 3^0 = 5$

$\Rightarrow \boxed{c_1 = 5.} \quad \text{--- (1)}$

Given $a_1 = 24 \Rightarrow (c_1 + c_2(1) + c_3(1)) 3^1 = 24$

$\Rightarrow \boxed{c_1 + c_2 + c_3 = 8} \quad \text{--- (2)}$

Given $a_2 = 117 \Rightarrow (c_1 + 2c_2 + 4c_3) 3^2 = 117.$

$\Rightarrow \boxed{c_1 + 2c_2 + 4c_3 = 13} \quad \text{--- (3)}$

Sub (1) in (2), $5 + c_2 + c_3 = 8 \Rightarrow c_2 + c_3 = 3.$

From (3) $5 + 2(c_2 + 2c_3) = 13 \Rightarrow c_3 = 3 - c_2$

$\Rightarrow c_2 + 2c_3 = \frac{13-5}{2}$

$\Rightarrow c_2 + 2c_3 = 4$

$\Rightarrow c_2 + 2(3 - c_2) = 4 \Rightarrow \boxed{c_2 = 2.}$

$\therefore c_3 = 3 - c_2 = 3 - 2 \Rightarrow \boxed{c_3 = 1}$

$\therefore$ The required solution = $\boxed{a_n = (5 + 2n + n^2) 3^n}$